



The on-premise versus cloud data storage debate is an ongoing, never ending one. When deciding on either of them, the security of the data is of utmost consideration. There are many myths and fears surrounding cloud security. This article seeks to dispel them by clearly outlining the safety and security in cloud storage, compared to on-premise storage.

The term cloud basically refers to 'anything' that is not on your computer or logical drives, and which can be accessed (read or written to) from your computer with the help of the Internet. This 'anything' on the cloud is stored, hosted and maintained in some data centre of usually a third-party storage provider. Some of the world's leading cloud services are Amazon Web Services (AWS), Microsoft Azure, Google Cloud, and so on.

As per the US based National Institute of Standards and Technology (NIST), there are three service models and four deployment models for cloud storage, as listed below.

Service models	Deployment models
1. Infrastructure-as-a-Service	1. Public cloud
2. Platform-as-a-Service	2. Private cloud
3. Software-as-a-Service	3. Hybrid cloud
	4. Community cloud

Nowadays, organisations are moving their infrastructure to the cloud on a large scale instead of setting up their own on-premise storage, which comes with huge installation and maintenance costs. The major benefits are high reliability, low maintenance and

optimised costs, as you pay only for what you use.

Cloud storage has a wide range of benefits but there are some concerns too. The major concerns arise with regard to security. What if a cloud storage provider's servers, which host its customers' sensitive data, are hacked into or compromised by some unauthorised person accessing them and, even worse, the clients are denied access to their own data?

Let's discuss cloud security so that it helps you stop worrying about it.

Cloud security vs on-premise security

Cloud security

To understand the security provided by the cloud, it is important to know how data is being stored in the cloud and how resources are provided to customers. This is done in two ways.

- **Shared infrastructure:** In this case, multiple customers are allotted resources from storage/drives of a single machine. This weakens the trust of customers, making them unwilling to store sensitive data. This can be conceived as renting different flats to different tenants